

MLB Batting Average

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COMP3125 Data Science

1. Introduction

Major League Baseball has had a shift in performance over the past few decades, in particular how batters perform at the plate. One of the most noticeable trends is the decline in league wide batting averages, which has reached one of the lowest averages in MLB history. This decline in batting averages has raised conversations about how pitching strategy, training, and technology have created an unbalanced skillset between offense and defense. Research suggests that the improvements of pitching like velocity, increased reliance on breaking balls, and advanced analytics guiding pitch selection have played a role in the drop of offensive output. Pitch velocities have increased since the early 2000s, making it more difficult for hitters to make consistent contact. At the same time, strikeouts rates have risen to historic highs while earned run averages (ERA) have shifted in ways that reflect pitching dominance. These different stats are important for understanding the correlation between pitching skills increasing and batting averages dropping.

This report will look into the relationship between these trends by examining pitch-level velocity over time, evaluating how strikeouts per game and ERA have changed in recent years, and analyzing how MLB's overall batting average has evolved.

2. Datasets

The dataset used for this analysis is from Baseball-Reference, a recognized and credible sports statistics provider frequently referenced by analysts, and MLB organizations. The dataset is updated and compiled from official MLB game records. The dataset is provided in tabular format and includes yearly aggregated statistics for all major league teams combined.

Year	MLB season year	Integer
BA	Batting Average(whole league)	Decimal
ERA	Earned Runs Average	
SO	Strikeouts(Whole season)	integer
SO/Game	Strikeouts per game	Decimal
HR	Home runs	integer
H	Hits	Integer

AB	At bats	integer
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For data cleaning I removed non numeric rows such as headers repeated within the table. I also handled the missing values by removing incomplete years.

3. Methodology

I am going to use descriptive methods and basic trend analysis to explore how MLB batting performance has changed in recent years in relation to pitching improvements.

A. Time-Series Trend Analysis

Time series analysis is used to observe changes in variables like batting average, strikeouts per game, ERA, and pitch velocity across multiple years. This method identifies upward or downward trends and shows periods of change. Some advantages of using a time series is that it is easy to interpret visually and statistically. It is also useful for identifying long-term changes. Some disadvantages are that it is sensitive to missing or inconsistent data points. I chose a time-series because this project focuses on how the statistics change over multiple years. I will use matplotlib and seaborn for line charts.

B. Correlation Analysis

Pearson correlation shows the relationship between pitch velocity, strikeouts, ERA, and batting average. Some advantages are that it expresses relationships between variables and is simple to compute and interpret. I chose a correlation analysis to help determine whether increasing pitch velocity or rising strikeouts correspond with decreases in batting average.